Pendulum

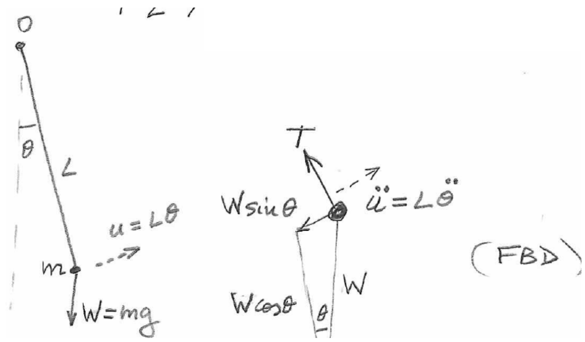

Figure 1: Pendulum geometry and free-body diagram.

For the pendulum, let $u = L\theta$ denote the tangential displacement and let $W = mg$. The sum of forces in the tangential direction is

$$\sum F_u = -W \sin\theta. \tag{1}$$

Newton's second law gives

$$\sum F_u = m\ddot{u}. \tag{1a}$$

Since $u = L\theta$,

$$-W \sin\theta = mL\ddot{\theta}. \tag{2}$$

For a small angle, $\sin\theta \approx \theta$ (radians), so

$$-W\theta = mL\ddot{\theta}, \tag{3}$$

and therefore the equation of motion is

$$mL\ddot{\theta} + W\theta = 0. \tag{4}$$

Using $W = mg$ in (4) gives

$$mL\ddot{\theta} + mg\theta = 0,$$

or the governing equation

$$\ddot{\theta} + \frac{g}{L}\theta = 0. \tag{5}$$

Assume a solution of the form

$$\theta = e^{st}, \qquad \ddot{\theta} = s^2 e^{st} = s^2\theta. \tag{6}$$

Substitution into (5) gives

$$\left(s^2 + \frac{g}{L}\right)\theta = 0, \tag{7}$$

with characteristic equation

$$s^2 + \frac{g}{L} = 0. \tag{8}$$

The roots are

$$s = \pm i\sqrt{\frac{g}{L}}. \tag{9}$$

Define the natural frequency in radians per second as

$$\omega_n = \sqrt{\frac{g}{L}}. \tag{10}$$

The solutions are

$$\begin{aligned} s_1 &= i\omega_n, \\ s_2 &= -i\omega_n. \end{aligned} \tag{11}$$

Thus,

$$\theta(t) = C_1^* e^{i\omega_n t} + C_2^* e^{-i\omega_n t}, \tag{12}$$

or, equivalently,

$$\theta(t) = A^* \cos\omega_n t + B^* \sin\omega_n t. \tag{13}$$

The frequency in hertz is

$$f_n = \frac{\omega_n}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{g}{L}}. \tag{14}$$

The period of oscillation is

$$\tau = \frac{1}{f_n} = 2\pi\sqrt{\frac{L}{g}}. \tag{15}$$

Particle motion

The particle displacement is

$$u = L\theta.$$

Consequently,

$$u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t}, \tag{16a}$$
$$u(t) = A\cos\omega_n t + B\sin\omega_n t, \tag{16b}$$
$$u(t) = C\sin(\omega_n t + \varphi). \tag{16c}$$

From complex exponentials to trigonometric functions

$$C_1 e^{i\alpha} + C_2 e^{-i\alpha} = A\cos\alpha + B\sin\alpha. \tag{12a}$$

or

$$C_1 e^{i\alpha} + C_2 e^{-i\alpha} = C\sin(\alpha + \varphi). \tag{12b}$$

Proof

Recall Euler's identity,

$$e^{i\alpha} = \cos\alpha + i\sin\alpha,$$

and hence

$$e^{-i\alpha} = \cos\alpha - i\sin\alpha.$$

Therefore,

$$\begin{aligned} C_1 e^{i\alpha} &= C_1\cos\alpha + iC_1\sin\alpha, \\ C_2 e^{-i\alpha} &= C_2\cos\alpha - iC_2\sin\alpha. \end{aligned}$$

Adding,

$$\begin{aligned} C_1 e^{i\alpha} + C_2 e^{-i\alpha} &= (C_1 + C_2)\cos\alpha + i(C_1 - C_2)\sin\alpha \\ &= A\cos\alpha + B\sin\alpha, \end{aligned}$$

where

$$A = C_1 + C_2, \qquad B = i(C_1 - C_2).$$

Let

$$C = \sqrt{A^2 + B^2}, \qquad \varphi = \tan^{-1}\frac{A}{B}.$$

Recall that

$$\sin(\alpha + \varphi) = \sin\varphi\cos\alpha + \cos\varphi\sin\alpha.$$

Hence

$$C\sin(\alpha + \varphi) = (C\sin\varphi)\cos\alpha + (C\cos\varphi)\sin\alpha.$$

Since

$$A^2 + B^2 = C^2\sin^2\varphi + C^2\cos^2\varphi = C^2,$$

we have $C = \sqrt{A^2 + B^2}$, and

$$\frac{A}{B} = \frac{C\sin\varphi}{C\cos\varphi} = \tan\varphi \quad\Longrightarrow\quad \varphi = \tan^{-1}\frac{A}{B}.$$

Thus,

$$C_1 e^{i\alpha} + C_2 e^{-i\alpha} = C\sin(\alpha + \varphi). \tag{12c}$$

Q.E.D.

Oscillatory motion

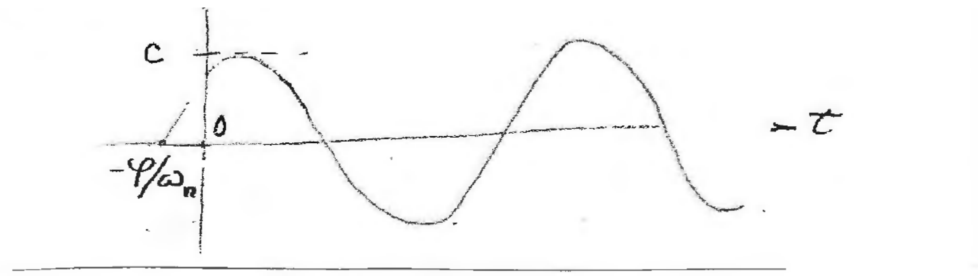

Figure 2: Oscillatory motion with amplitude and phase.

The displacement may be written in either of the equivalent forms

$$u(t) = C\sin(\omega_n t + \varphi). \tag{13a}$$

$$u(t) = A\cos\omega_n t + B\sin\omega_n t. \tag{13b}$$

The constants A, B or C, φ must be determined from the initial conditions.

Initial-value problem

Assume that the initial displacement, u_0, and the initial velocity, $\dot{u}_0$, are known. Using Eq. (13b), we write

$$\begin{aligned} u_0 = u(0) = {}& (A\cos\omega_n t + B\sin\omega_n t)|_{t=0} = A, \\ \dot{u}_0 = \dot{u}(0) = {}& (-\omega_n A\sin\omega_n t + \omega_n B\cos\omega_n t)|_{t=0} = \omega_n B. \end{aligned} \tag{26}$$

Solving Eq. (26) for A and B yields

$$\begin{aligned} A &= u_0, \\ B &= \frac{\dot{u}_0}{\omega_n}. \end{aligned} \tag{27}$$

Substitution of Eq. (27) into Eq. (13b) gives the general solution of undamped free vibrations with initial displacement, u_0, and initial velocity, $\dot{u}_0$, in the form

$$u(t) = u_0\cos\omega_n t + \frac{\dot{u}_0}{\omega_n}\sin\omega_n t. \tag{28}$$

Complex-exponential form

The solution can also be written as

$$u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t}. \tag{1}$$

The initial-value conditions are

$$\begin{aligned} u(0) &= u_0, \\ \dot{u}(0) &= \dot{u}_0. \end{aligned} \tag{2}$$

Differentiating Eq. (1) gives

$$\dot{u}(t) = i\omega_n C_1 e^{i\omega_n t} - i\omega_n C_2 e^{-i\omega_n t}. \tag{3}$$

Thus,

$$\begin{aligned} u(0) &= C_1 + C_2, \\ \dot{u}(0) &= i\omega_n C_1 - i\omega_n C_2. \end{aligned} \tag{4}$$

Using Eq. (4) with the initial conditions gives

$$\begin{aligned} C_1 + C_2 &= u_0, \\ i\omega_n C_1 - i\omega_n C_2 &= \dot{u}_0. \end{aligned} \tag{5}$$

Adding and subtracting these equations yields

$$\begin{aligned} 2i\omega_n C_1 &= i\omega_n u_0 + \dot{u}_0, \\ -2i\omega_n C_2 &= -i\omega_n u_0 + \dot{u}_0, \end{aligned} \tag{6}$$

and therefore

$$\begin{aligned} C_1 &= \frac{u_0}{2} + \frac{\dot{u}_0}{2i\omega_n}, \\ C_2 &= \frac{u_0}{2} - \frac{\dot{u}_0}{2i\omega_n}. \end{aligned} \tag{7}$$

Substitution into Eq. (1) gives

$$\begin{aligned} u(t) &= \left(\frac{u_0}{2} + \frac{\dot{u}_0}{2i\omega_n}\right) e^{i\omega_n t} + \left(\frac{u_0}{2} - \frac{\dot{u}_0}{2i\omega_n}\right) e^{-i\omega_n t} \\ &= u_0 \frac{e^{i\omega_n t} + e^{-i\omega_n t}}{2} + \frac{\dot{u}_0}{\omega_n} \frac{e^{i\omega_n t} - e^{-i\omega_n t}}{2i} \\ &= u_0 \cos \omega_n t + \frac{\dot{u}_0}{\omega_n} \sin \omega_n t, \end{aligned}$$

as before.

Harmonic oscillation

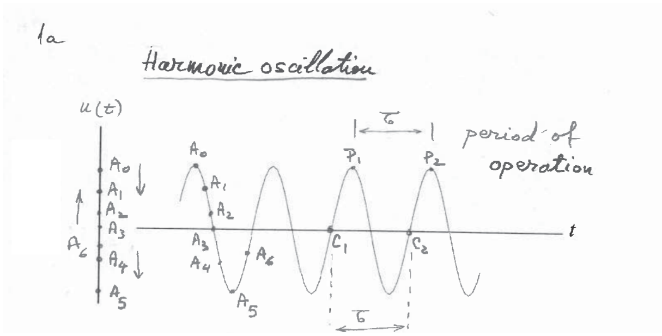

Figure 3: Harmonic oscillation, peaks, zero crossings, and period.

The particle vibrates up and down, going through positions A_0, A_1, A_2, A_3, A_4, A_5, and A_6. Graphically, this is represented as an oscillatory curve which has peaks and valleys.

The period of oscillation, τ, is the distance in time between two consecutive peaks, P_1 and P_2, or two consecutive zero crossings with positive slope, C_1 and C_2.

The frequency, f, is how often the particle passes through the same location. Frequency f is the inverse of period τ,

$$f = \frac{1}{\tau}\ \text{Hz}.$$

The angular frequency ω is

$$\omega = 2\pi f\ \text{rad/s}.$$

1-dof VIBRATION

1 Spring–Mass Vibrating Systems

Consider a vibrating system consisting of a mass m and a spring of stiffness k as shown in Figure 1.

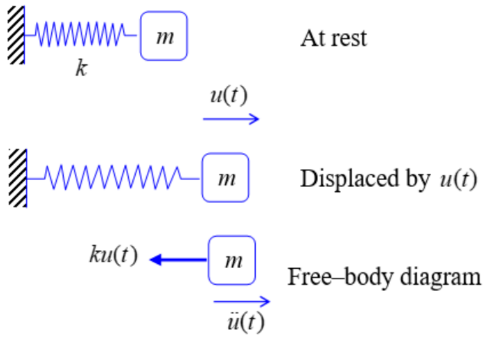

Figure 1: Spring-mass vibration system

1.1 Free Vibration of Spring–Mass Systems

Assume the mass m is displaced from the equilibrium position by a time-dependent displacement $u(t)$.

1.1.1 Free-Body Diagram

As shown in Figure 1, the free body diagram (FBD) of the mass m gives the sum of x-direction forces as

$$\sum F_x = -ku(t) \tag{1}$$

where $ku(t)$ is the reaction force from the spring.

1.1.2 Newton's Law of Motion

Newton's law of motion (NLM) applied to the mass m is stated as

$$\sum F_x = m\ddot{u}(t). \tag{2}$$

1.1.3 Equation of Motion

Substitution of Eq. (1) into Eq. (2) yields

$$m\ddot{u}(t) = -ku(t). \tag{3}$$

Upon rearrangement, Eq. (3) becomes the equation of motion (EOM), i.e.,

$$m\ddot{u}(t) + ku(t) = 0. \tag{4}$$

Eq. (4) is a linear homogeneous ordinary differential equation (ODE).

1.1.4 Solution of Linear ODEs

The solution of linear homogeneous ODEs is sought in the form e^{st}.

$$u(t) = e^{st}. \tag{5}$$

Upon differentiation, Eq. (5) gives

$$\begin{aligned} \dot{u}(t) &= se^{st} = su(t), \\ \ddot{u}(t) &= s^2e^{st} = s^2u(t). \end{aligned} \tag{6}$$

Substitution of Eqs. (5), (6) into Eq. (4) yields

$$ms^2u(t) + ku(t) = 0. \tag{7}$$

After factoring out $u(t)$, Eq. (7) becomes

$$(ms^2 + k)u(t) = 0. \tag{8}$$

A possible solution of Eq. (8) is $u(t) = 0$. However, this solution is trivial and of no interest to us. We are interested in nontrivial solutions $u(t) \neq 0$. To obtain a nontrivial solution $u(t) \neq 0$, one needs to find the values of s that make zero the factor multiplying $u(t)$ in Eq. (8), i.e., to have $(ms^2 + k) = 0$.

1.1.5 Characteristic Equation

It was shown in Section 1.1.4 that a nontrivial solution $u(t) \neq 0$ of Eq. (8) can only be obtained if the factor multiplying $u(t)$ is zero, i.e.,

$$ms^2 + k = 0. \tag{9}$$

Divide Eq. (9) by m to get what is called the characteristic equation, i.e.,

$$s^2 + \frac{k}{m} = 0 \qquad \text{(characteristic equation)}. \tag{10}$$

Solution of the characteristic equation will give us the values of s that permit nontrivial solution $u(t) \neq 0$ of Eq. (8). Upon solution, Eq. (10) gives

$$s^2 = -\frac{k}{m}. \tag{11}$$

Denote

$$\omega_n^2 = \frac{k}{m}, \qquad \omega_n = \sqrt{\frac{k}{m}} \quad \text{(natural frequency)}. \tag{12}$$

The natural frequency ω_n is expressed in rad/sec. Substitution of Eq. (12) into Eq. (11) yields

$$s^2 = -\omega_n^2. \tag{13}$$

Eq. (13) has two imaginary roots

$$s_1 = +i\omega_n, \qquad s_2 = -i\omega_n. \tag{14}$$

1.1.6 General Solution

Both $s_1 = i\omega_n$ and $s_2 = -i\omega_n$ yield solutions of the ODE Eq. (4). These solutions are in the form of complex exponentials $e^{i\omega_n t}$ and $e^{-i\omega_n t}$, respectively. Hence, the general solution of the linear ODE Eq. (4) is a linear superposition of the two individual solutions, i.e.,

$$u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t} \tag{15}$$

where C_1 and C_2 are arbitrary constants that need to be determined from the initial conditions.
Recall Euler's identity

$$e^{i\alpha} = \cos\alpha + i\sin\alpha \tag{16}$$

and hence

$$e^{-i\alpha} = \cos\alpha - i\sin\alpha. \tag{17}$$

Eqs. (16), (17) can be used to rewrite Eq. (15) in trigonometric form, i.e.,

$$u(t) = A\cos\omega_n t + B\sin\omega_n t. \tag{18}$$

where the constants A, B and C_1, C_2 are interconnected as

$$A = C_1 + C_2, \tag{19}$$

$$B = i(C_1 - C_2). \tag{20}$$

Further processing allows us to write Eq. (18) as

$$u(t) = C\sin(\omega_n t + \varphi) \tag{21}$$

where

$$C = \sqrt{A^2 + B^2}, \qquad \varphi = \tan^{-1}\frac{A}{B}. \tag{22}$$

1.1.7 Initial Value Problem (IVP)

The initial value problem (IVP) consists in finding the general-solution constants in terms of the initial conditions of the vibrating system, i.e., initial displacement u_0 and initial velocity $\dot{u}_0$. The purpose is to find the constants A, B in terms of known u_0 and $\dot{u}_0$. Using Eq. (18), we write

$$u_0 = u(0) = A\cos\omega_n t + B\sin\omega_n t\big|_{t=0} = A,$$
$$\dot{u}_0 = \dot{u}(0) = -\omega_n A\sin\omega_n t + \omega_n B\cos\omega_n t\big|_{t=0} = \omega_n B. \tag{23}$$

Solving Eq. (23) for A and B yields

$$A = u_0,$$
$$B = \frac{\dot{u}_0}{\omega_n}. \tag{24}$$

Substitution of Eq. (24) into Eq. (18) gives the general solution of undamped free vibrations with initial displacement, u_0, and initial velocity, $\dot{u}_0$, in the form

$$u(t) = u_0\cos\omega_n t + \frac{\dot{u}_0}{\omega_n}\sin\omega_n t. \tag{25}$$

1.1.8 Vertical spring–mass analysis

So far, we have assumed the mass–spring vibrating system is horizontal. However, many spring–mass vibrating systems are vertical. In this section, we show that a vertical spring–mass obeys the same differential equation as a horizontal one, which means that the analysis outlined so far applies equally to both situations.

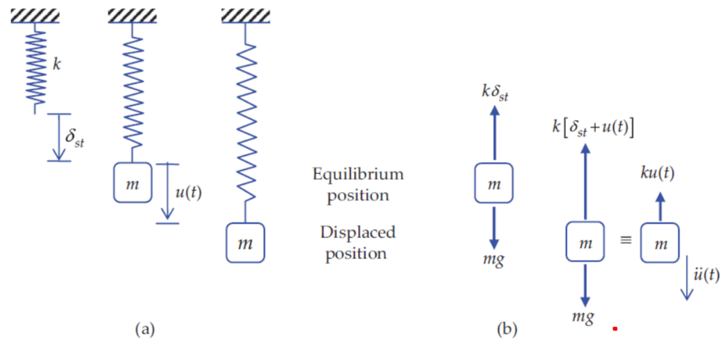

Figure 2: Free vibration of a vertical spring-mass system: (a) physical mechanism; (b) free-body diagram

Consider a vertical spring–mass vibration system consisting of a particle of mass m hanging on an elastic spring of stiffness k as shown in Figure 2.

1.1.8.1 Static analysis

Initially, the mass is at rest and in a state of equilibrium with the static tension in the spring (Figure 2a). In this equilibrium state, the weight $W = mg$ is balanced by the static tension in the spring $k\delta_{\text{st}}$, where δ_{st} is the static displacement of the spring. The free body diagram of the mass in equilibrium condition is shown in Figure 2b. Force-balance analysis in the vertical direction gives

$$\sum F_y = mg - k\delta_{\text{st}} = 0 \qquad \text{(static equilibrium)} \tag{26}$$

Solution of Eq. (26) yields the static displacement δ_{st} as

$$\delta_{\text{st}} = \frac{mg}{k} \qquad \text{(static displacement)} \tag{27}$$

1.1.8.2 Vibration analysis

If the mass is displaced from this equilibrium state and then let free, it will oscillate up and down about the equilibrium position, i.e., it will experience a state of vibration with time-dependent displacement $u(t)$. The free body diagram of the mass in the displaced position gives the sum of forces in the vertical direction as

$$\sum F_y = mg - k\left[\delta_{\text{st}} + u(t)\right] \qquad \text{(vibration motion)} \tag{28}$$

Expansion of Eq. (28) gives

$$\sum F_y = (mg - k\delta_{\text{st}}) - ku(t) \qquad \text{(vibration motion)} \tag{29}$$

However, $(mg - k\delta_{\text{st}}) = 0$ as indicated by the equilibrium analysis in Eq. (26). Hence, the sum of forces acting on the vibrating mass is simply

$$\sum F_y = -ku(t) \qquad \text{(sum of forces for vibration analysis)} \tag{30}$$

Newton's law of motion (NLM) for the vertical direction is stated as

$$\sum F_y = m\ddot{u} \qquad \text{(NLM)} \tag{31}$$

Substitution of Eq. (30) into Eq. (31) gives, upon rearrangement,

$$m\ddot{u}(t) + ku(t) = 0. \tag{32}$$

Eq. (32) is the same as Eq. (4) developed previously for the horizontal spring–mass vibrating system. Its general solution is that given by Eq. (21), i.e.,

$$u(t) = C\sin(\omega_n t + \varphi). \tag{21}$$

A rendition of the vertical vibration motion is depicted in Figure 3.

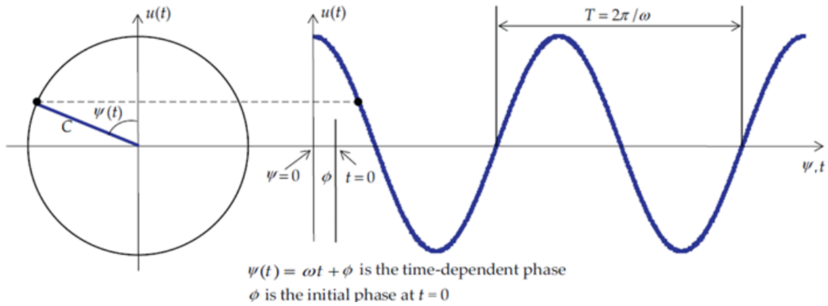

Figure 3: Schematic representation of an oscillatory motion of amplitude C, angular frequency ω, and initial phase ϕ

Note that the above analysis proved that the vibration equation is not modified if static gravity is also included in the analysis. This means that the vibration motion is not affected by the presence of a static force field such as gravity. The analysis can be easily extended to a static force field that is not parallel with the vibration motion, but it has an arbitrary orientation.